MA1513 Weekly Self Practice Set 1 Answer

1. (a) Write down the original linear system with the following augmented matrix. (Use variables x_1, x_2, x_3, x_4)

$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 2 & 5 & -4 & 2 & 15 \\ 3 & 5 & -1 & 3 & 20 \end{array}\right)$$

Answer

$$\left\{\begin{array}{ccccccccc} x_1 & & & + & 3x_3 & + & x_4 & = & 5 \\ 2x_1 & + & 5x_2 & - & 4x_3 & + & 2x_4 & = & 15 \\ 3x_1 & + & 5x_2 & - & x_3 & + & 3x_4 & = & 20 \end{array}\right.$$

(b) Use Gaussian elimination to reduce the augmented matrix in part (a) to the reduced row echelon form

$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

using exactly four elementary row operations. Write down the elementary row operation that you use in each step clearly.

Answer

$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 2 & 5 & -4 & 2 & 15 \\ 3 & 5 & -1 & 3 & 20 \end{array}\right) \xrightarrow{R_2-2R_1} \left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 0 & 5 & -10 & 0 & 5 \\ 3 & 5 & -1 & 3 & 20 \end{array}\right)$$

$$\xrightarrow{R_3-3R_1} \left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 0 & 5 & -10 & 0 & 5 \\ 0 & 5 & -10 & 0 & 5 \end{array}\right) \xrightarrow{R_3-R_2} \left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 0 & 5 & -10 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow{\frac{1}{5}R_2} \left(\begin{array}{cccc|c} 1 & 0 & 3 & 1 & 5 \\ 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

(c) Write down the general solution of the linear system in part (a).

Answer

Substitute x_3 and x_4 with parameters: $x_3 = t,\ x_4 = s$. Then the general solution is given by

$$\left\{\begin{array}{lcl} x_1 & = & 5 - 3t - s \\ x_2 & = & 1 + 2t \\ x_3 & = & t \\ x_4 & = & s \end{array}\right.$$

2. (a) Write the linear system

$$\begin{cases} 3x & + & 2y & = & 6 \\ 2x & + & 2y & = & 8 \end{cases}$$

in matrix equation form $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.

Answer

$$\begin{pmatrix} 3 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}.$$

(b) Find the inverse matrix $\boldsymbol{A}^{-1}$ of $\boldsymbol{A}$ in part (a) by hand.

Answer

$$\boldsymbol{A}^{-1} = \frac{1}{2} \begin{pmatrix} 2 & -2 \\ -2 & 3 \end{pmatrix}$$

(c) Use the answer in (b) to find the solution of the matrix equation form $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ in (a). Show your working, and give your answer of $\boldsymbol{x}$ as a column matrix.

Answer

The solution is given by

$$\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b} = \frac{1}{2} \begin{pmatrix} 2 & -2 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 6 \\ 8 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}.$$